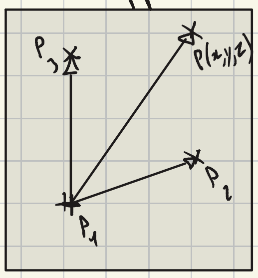
Lecture 3: Matrices

$\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$; $\vec{A}\times\vec{A} = 0$

Application: Let's say we want to find the equation of a plane containing P_1, P_2, P_3 which means, finding a condition on (x,y,z) telling us wether P is in plane or not



To find out if it's a plane or not we could ask ourselves if the parallelepiped they form is flattened or not; if flat, then it's a plane if not, then the parallelepiped has a volume and it's not a plane.

In other words, if it's a plane, $\det\left(\overrightarrow{P_1P_2} ; \overrightarrow{P_1P} ; \overrightarrow{P_1P_3}\right) = 0$

Other solution: P is in the plane if: $\overrightarrow{P_1P} \perp \vec{N}$ (some vector $\perp$ to the plane, "normal vector")
$\Leftrightarrow \overrightarrow{P_1P}\cdot\vec{N} = 0$

How to find $\vec{N} \perp$ plane ? $\vec{N} = \overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}$

So : $\overrightarrow{P_1P}\cdot\vec{N} = \underbrace{\overrightarrow{P_1P}\cdot\left(\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}\right)}_{\text{triple product = det}} = 0$

Example:
$$\begin{cases} u_1 = 2x_1 + 3x_2 + 3x_3 \\ u_2 = 2x_1 + 4x_2 + 5x_3 \\ u_3 = x_1 + x_2 + 2x_3 \end{cases} \Leftrightarrow \underbrace{\begin{bmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}}_{U}$$

For a matrix product of M^1_{rp} M^2_{ks} you must have $p = k$ (nb of columns of M^1 = nb of rows of M^2)

What AB represents: do transformation B, then transformation A.

$A(BX) = (AB)X$

Identity Matrix: $I_{3\times3}$ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Example: in the plane, rotation by $90°$ counterclockwise is made by the matrix: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Rotation of $90°$ of $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$

Inverse Matrix: A needs to be a square matrix

Let inverse of Matrix $A = M$
then, $AM = I$ and $MA = I$

Solution to $AX = B$ is $X = A^{-1}B$

$AX = B \Leftrightarrow \underbrace{A^{-1}A}_{I}X = A^{-1}B \Leftrightarrow X = A^{-1}B$

Formula: $A^{-1} = \dfrac{1}{\det(A)}\overbrace{\operatorname{adj}(A)}^{\text{"adjoint"}}$

Steps on a 3x3 example: $A = \begin{bmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{bmatrix}$ M_1 M_3

① Minors $\begin{pmatrix} 3 & -1 & -2 \\ 3 & 1 & -1 \\ 3 & 4 & 2 \end{pmatrix}$

$\det\begin{pmatrix} 4 & 5 \\ 1 & 2 \end{pmatrix} = M_1$ $\det\begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix} = M_3$

② Cofactors:

flip signs: $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} = \begin{pmatrix} 3 & +1 & -2 \\ -3 & 1 & +1 \\ 3 & -4 & 2 \end{pmatrix}$

③ Transpose: (switch rows & columns)

$\begin{pmatrix} 3 & -3 & 3 \\ 1 & 1 & -4 \\ -2 & 1 & 2 \end{pmatrix} = \operatorname{adj}(A)$

④ Divide by det $\det(A) = 3$

$A^{-1} = \dfrac{1}{3}\begin{bmatrix} 3 & -3 & 3 \\ 1 & 1 & -4 \\ -2 & 1 & 2 \end{bmatrix}$